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Review

Comprehensive review on women safety technologies: Indian perspective

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ABSTRACT

Recently, there has been a concerning escalation in crimes against women, evidenced by a nearly doubled rate of reported cases from approximately 371,503 in 2020 to exceeding 445,256 in 2022. Shockingly, government data reveals a reported rape every 15 min, indicating a dire lack of safety for women and girls, even within their own homes or communities. Moreover, it conducts a thorough evaluation of Women Safety Technologies (WSTs), such as mobile applications, safety wearables, Global Positioning System (GPS) tracking, public surveillance systems, and Internet of Things (IoT)-based smart tracking devices, aimed at mitigating these issues. Thus, introducing the "SIRIP (Search, Inclusion, Review, Interpretation, and Presentation)" search strategy to enhance evaluation effectiveness is essential. To address these shortcomings and enhance overall safety measures, the paper presents a taxonomy elucidating the limitations of current WST and advocates for technological advancements. Reviewed standardized research papers from 2018 to 2025 underscore various technologies, algorithm applications, and sensor integration, alongside discussing their outcomes. A four-level taxonomy of women's safety devices has been delineated, emphasizing user-friendly features to improve women's safety.

1. Introduction

Examining crimes against women in India under the Indian Penal Code reveals concerning statistics (Ministry of External Affairs, 2023). According to the National Crime Records Bureau (NCRB), the total number of crimes against women rose from 371,503 cases in 2020 to 445,256 in 2022, reflecting a steady and troubling increase in reported incidents over just two years (NCRB, 2023; PIB, 2022). "Assault to outrage her modesty" constitutes 20.8 %, "kidnapping and abduction of women" constitutes 17.6 %, and "rape" constitutes 7.4 % (Ramachandran, 2022). Survivors often face long-term trauma, like depression, anxiety, Post-Traumatic Stress Disorder (PTSD), and suicidal ideation, thereby severely affecting their quality of life. Many are stigmatized or blamed, leading to social isolation and emotional distress. Psychological effects on survivors can range from immediate physical injuries and sexually transmitted infections to long-term mental health issues like depression, PTSD, and substance abuse (Airaoje et al., 2025). Research evidence now clearly exhibits a direct link between women's experiences of domestic violence and heightened rates of depression, trauma symptoms, and self-harm (Jain & Pandey, 2025). From a criminological perspective, WSTs can be situated within the framework of situational crime prevention. At a national level, the economic cost of such violence is substantial—estimated in some studies to be as high as 1–2 % of a country's Gross Domestic Product (GDP). Street harassment is widespread, with approximately 80 % of women experiencing it occasionally (Ministry of Home Affairs, 2022). Statistics from 2018 highlight a significant number of adult women and girls trafficked for sexual exploitation (Statista, 2024). Various technologies have been deployed to mitigate threats against women in public spaces, including mobile applications, wearable devices, embedded systems, drones, and surveillance systems (The New India Express, 2022).

1.1. Evolution of WST

This section maps the progression of key developments, especially within the Indian context, showcasing how innovation has responded to emerging safety challenges. The 2000s ushered in the advent of mobile technology, drastically improving access to emergency services and personal networks through mobile phones and marking a pivotal shift in personal safety dynamics (Teodorescu et al., 2023). The 2010s saw the rise of wearable technology, including smartwatches, fitness bands, and smart jewelry, which were designed with both health and safety in mind. Simultaneously, public surveillance infrastructures expanded, transitioning from targeted security setups to widespread urban monitoring,

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which enhances both deterrence and evidence-gathering capabilities (Nishnianidze, 2023). This decade also witnessed the emergence of the IoT and smart devices, where interconnected systems, ranging from smart locks to lighting, played an increasingly central role in public and domestic safety (Allanki Sanyasi Rao et al., 2023). Artificial Intelligence (AI) and big data technologies aim to anticipate threats and optimize preventive responses (Kalla & Smith, 2024). As new tools continue to emerge, their societal acceptance and ethical governance are critical in determining their long-term impact (Bhat, 2019).

1.2. AI-driven predictive safety systems

Safety-critical environments demand operational precision and high-reliability mechanisms, often subject to formal certification and oversight (Wang & Chung, 2022). Predictive frameworks use data modeling, machine-driven pattern recognition, sensor-based inputs, and simulation tools to manage energy use, streamline operations, and prevent environmental or personal harm (Park & Kang, 2024). Applications with behavioral analysis engines are being developed to curb harassment in public spaces, reduce incidents of domestic violence, and improve response time during emergencies (Jewani et al., 2024). A growing number of software platforms focus on creating safe environments through advanced detection and response capabilities. These tools aim to reduce instances of assault and harassment by offering predictive alerts and support networks (Ponnusamy et al., 2024). One such example is SecureMe, a safety platform integrating smart sensors and continuous data processing to deliver personalized alerts and connect users to emergency services and community support in real time (Kajale et al., 2025). Surveillance tools, such as Closed-Circuit Television (CCTV) cameras, placed in public areas and transportation hubs have proven effective as both crime deterrents and sources of post-incident evidence (Welsh et al., 2018).

Wearable devices primarily encompass wearable gadgets like keychains, bands, watches (Khandelwal et al., 2018), shoes, and jackets embedded with sensors. Also, the NCRB (Press Information Bureau, 2022), World Health Organization (WHO) (World Health Organization, 2024), and UN Women (UN Women Asia and the Pacific, 2023) reports present a worrisome picture of women's safety in India and around the world. The literature review's primary objective is to spotlight the shortcomings of existing apps and introduced devices. However, their vulnerability is further exacerbated by limited access to economic resources and weaker social standing, which significantly increases their risk during emergencies and crises (Ullah, 2018). This review, while comprehensive in examining women's safety technologies within the Indian context, has certain limitations in its scope that must be acknowledged. Firstly, the focus on India, though intentional and necessary for addressing region-specific challenges, may restrict the findings' generalizability to other countries or cultural contexts. Secondly, the review emphasizes technological interventions, such as mobile apps, wearable devices, and surveillance systems, while largely excluding non-technological or community-based safety strategies like education, legal reforms, or grassroots advocacy, which are also crucial components of the broader safety ecosystem. Additionally, only English-language and peer-reviewed sources were included, potentially overlooking valuable grey literature and regionally published studies in Indian languages. Nirbhaya Fund was specifically designed to finance projects that were aimed at enhancing women's safety, including the implementation of panic buttons on mobile phones, GPS tracking in public transport, and city-wide surveillance systems (Gupta, 2023). Additionally, schemes like the Safe City Project under the Ministry of Home Affairs and the One Stop Centre Scheme by the Ministry of Women and Child Development have promoted the integration of digital tools with legal and psychosocial support services. These policies reflect a growing recognition of the role of technology in gender-centered violence prevention, while also emphasizing the importance of institutional coordination, local context, and sustained funding (Hawken et al., 2020). Key legal reforms like the Criminal Law (Amendment) Act (2013), introduced in the wake of the Nirbhaya case, expanded the definitions of sexual offenses and established stricter punishments for gender-based violence, creating a legal foundation for responsive safety interventions (Gaur & Chauhan, 2024). The integration of legal studies into this review underscores the importance of a robust legal ecosystem to not only mandate technological safeguards but also ensure ethical deployment, user privacy, and institutional responsibility in promoting women's safety (Ikbal, 2023). Tools, such as CCTV cameras and GPS tracking, create risks related to privacy violations, unauthorized data sharing, and potential misuse by third parties or even abusers (Osipova & Hornecker, 2023). Cost and affordability remain significant issues as many devices and mobile applications require smartphones, internet connectivity, or subscriptions, which are not accessible to all segments of society, especially in rural and low-income populations. Digital literacy and accessibility also pose challenges as women with limited technical knowledge may struggle to effectively use or trust such technologies (Sambasivan et al., 2019). Furthermore, cultural and societal norms often discourage women from using safety tools that may be perceived as undermining family or community authority, particularly in conservative settings. In this literature review, a total of 78 papers were reviewed from 2018 to 2025, and ultimately, 30 of them were finalized based on certain criteria using the SiRip technique, which was a strategy for data inclusion and exclusion.

The subsequent sections of this paper are organized as follows: Section II delves into related works, while Section III outlines the research methodology adopted in this review. Sections IV and V present the analysis and results representation, including the proposed taxonomy, along with a discussion on unresolved issues and challenges that can guide future perspectives in this domain, followed by the conclusion in Section VI.

2. Related work

A survey study by Muhammad Shoaib Farooq focused on IoT devices that clearly classified the use of devices and their shortcoming. On the other hand, women who were less familiar with IoT devices or Machine Learning (ML) algorithms might face challenges in effectively using these tools. The usage of the GPS and Global System for Mobile Communication (GSM) or Raspberry Pi (Yaswanth et al., 2020) facilitated location tracking and sending distress messages to the woman's guardian. Jasmeeth Kaur et al.'s survey focused on embedded systems with ML/DL algorithms, emphasizing human body sensor detection and addressing prior limitations. Interconnected devices enabled real-time monitoring and rapid threat response. Meanwhile, U Y Reddy's study emphasized surveillance drones' pivotal role, addressing their limitations and advocating for continuous monitoring. R.R. Khandakor's research underscored the significance of mobile apps in women's safety. (Theivanathan et al., 2020) explained the Women's Safety Patrolling Drone using advanced technologies. The drone rapidly arrived at the location to protect the women. In this way, threats against women were prevented. (Stanney et al., 2025) described another advanced technology (virtual reality) for women's safety.

Women's safety in India is primarily concerned with challenges like harassment, crime against women at the workplace, domestic violence, and inadequate public security, which necessitate widespread measures for ensuring a safer environment for women. The main answer to the issue is that women should be allocated a well-being convenience device that guarantees their security (Malaj, 2023). The risk detection methodology centered on AI is trained on datasets mimicking actual emergency and non-emergency situations, so that the system can detect risky environments or actions in real time (Khan et al., 2025). Despite the widespread availability of communication tools, the response time for assistance can still be delayed, putting women at risk. To tackle the delay, a system has been designed specifically for offering real-time support in emergencies, offering women a sense of security and the

assurance that help is just a button press away (Choudhary & Gupta, 2024). By incorporating GPS and GSM technologies, the device aims to detect potential threats automatically, forward alerts to predefined contacts, and relay the user's location coordinates to facilitate swift and targeted responses from dear ones and law enforcement (Ingole et al., 2024). Table 1.1 explores the overview of prominently known mobile apps. A safety device may help women when there is an emergency. GPS helps women to move freely wherever they want (Jesudoss et al., 2018).

The credibility and trustworthiness of women's safety applications, such as Safetipin, VithU, bSafe, My Safetipin, Raksha, Circle of 6, and Life360, vary based on factors like developer transparency, user reviews, data privacy policies, and integration with emergency services. Safetipin and My Safetipin, developed in collaboration with urban safety experts and Non-Governmental Organizations (NGOs), are generally regarded as reliable due to their data-driven approach to mapping unsafe areas and community-based input. VithU, backed by a prominent media house, gains visibility through celebrity endorsements but has received mixed reviews regarding consistent functionality. bSafe and Life360, developed internationally, are often praised for real-time location tracking and emergency alerts, although concerns about data sharing and user surveillance have been raised.

2.1. Pre-digital foundations of women's safety devices

Early personal safety tools included safety whistles, self-defense keychains, personal alarms, and pepper sprays; many of which gained widespread popularity in the 1970s and 1980s as public awareness of women's vulnerability in urban environments grew. In India, the Nirbhaya Fund, established in 2013 in response to the Delhi gang rape case, is a dedicated corpus aimed at implementing initiatives to augment women's safety in public spaces (Singh & Singh, 2020).

3. Research methodology

In this research review, a novel search strategy method known as SIRIP [3.1] is employed. This innovative method, **SIRIP,** offers an effective framework for systematically crafting and implementing search strategies in research endeavors. It stands for "Select, Input, Run, Investigate, and Produce". The SIRIP method is used to explore women's safety technology, particularly in research endeavors. The SIRIP method, while systematic and structured, is widely used in current review papers, thus representing a conventional approach to literature synthesis. Publication bias may be present as studies with positive or innovative findings are more likely to be published, potentially overlooking unsuccessful or underreported initiatives. Language bias is another limitation, as the review includes only English-language sources, thereby possibly excluding relevant studies or government reports published in regional Indian languages. Fig. 1 explores the SIRIP scheme.

3.1. SIRIP scheme

A. **Select (Database Selection):**

Research initiated by Research Questions [RQs] and Research Motive [RM] is shown in Table 3.1A. Here, the sources used are IEEE, ACM Digital, Research Gate, Science Direct, Springer, and Elsevier.

B. **Input (Identifying keywords):**

A comprehensive list of keywords, phrases, and terminology relevant to women's safety technology is identified and compiled. As shown in Table 3.1B, keywords can be divided into primary, secondary, and tertiary. The contents related to primary, secondary, and tertiary keywords are shown in Table 3.1B.

C. **Run (Forming search queries):**

Selected keywords from the above two steps and phrases are constructed as search queries. Boolean operators (AND, OR, and NOT) are applied to combine these terms effectively. The search queries across the chosen databases or platforms are executed to retrieve relevant literature, articles, patents, or research papers, as shown in citation 1 and Table 3.1C.

3.2. Citation 1: Combination of all 3 keywords to form a search thread

∀ Primary keyword ∧ ∀ Secondary keyword V Tertiary keyword ∧Narrow
(∀ Devices V Technology V IoT devices V wearables V Gadgets V Surveillance System V mobile app V Drone) ∧ (Women's safety V women's security) V (Physical violence V Rape V Threat V Street abusing V kidnapping V human trafficking)∧**India**

A Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart is essential for systematically gathering articles for a comprehensive review on women's safety technologies, particularly from an Indian perspective. Fig. 2 explores the PRISMA flowchart.

D. **Investigate:**

Among 78 papers, 6 papers are excluded by applying SIRIP filters and limitations, as shown in Table 3.1D. Only papers published after 2018 are included, while those predating 2018 are disregarded.

For the comprehensive review on WSTs from an Indian perspective, a structured search strategy is employed using electronic databases, such as IEEE Xplore, Scopus, PubMed, Google Scholar, and ScienceDirect. Keywords and Boolean operators, such as "women safety," "technology," "India," "mobile applications," "wearable devices," "surveillance systems," and "gender-based violence," are used to detect pertinent studies. Inclusion criteria concentrate on peer-reviewed articles, conference papers, government reports, and case studies, specifically addressing WSTs implemented or studied within the Indian context. Studies that lack a technological focus do not pertain to India, or are purely opinion-based without empirical or technical grounding, are excluded.

E. **Produce:**

Produce is the final step of the RM process, where the ultimate

Table 1.1
An overview of prominently known mobile apps.

Mobile apps	Descriptions	Links
VithU	Sends alert messages to emergency contacts with the user's location by pressing the power button twice in case of danger.	(Ashok et al., 2022)
bSafe	Offers an SOS button, live GPS tracking, fake calls for quick exits, and a 'Follow Me' feature for friends to track their location.	(Poorna et al., 2020)
My Safetipin	Safetipin offers safety scores for areas and allows users to mark areas as 'unsafe', providing insights to others.	(Manazir et al., 2019)
Raksha	Sends SOS messages to pre-added contacts with location details when activated.	(Bhadre et al., 2024)
Circle of 6	Allows users to select six trusted contacts for immediate emergency contact.	(Paramanathan, 2021)
Life360	While not specifically for women's safety, it provides location tracking and family safety features.	(Long et al., 2023)

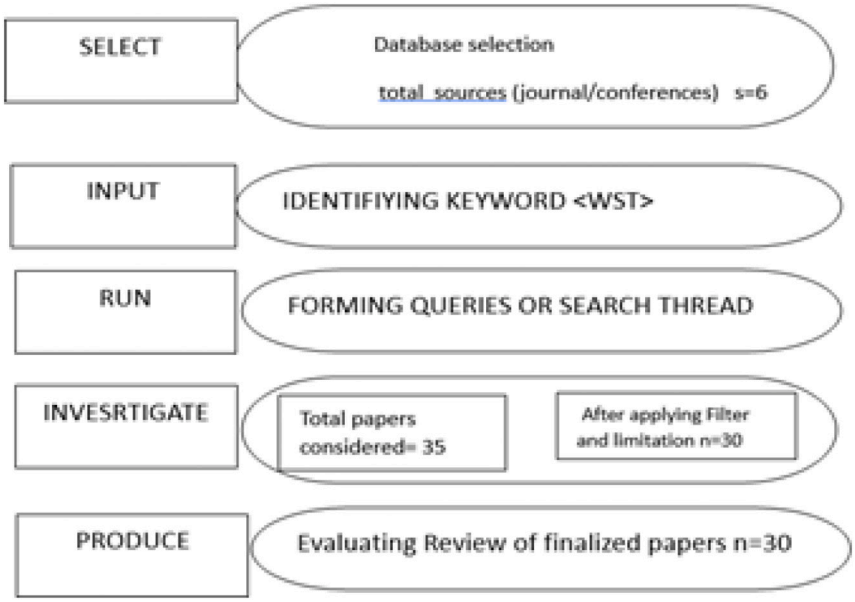

Fig. 1. SIRIP scheme.

Table 3.1A
Table showing research question and research motive.

RQ 1: What are the different WSTs available?
RM 1. a: To understand various technologies involved in women's safety.
RM 1.b: Creates Awareness among women regarding the availability of technology for their safety.
RQ 2: What are the prominent features used in developing these WSTs?
RM 2: To understand the different sensors or algorithms used in this technology.
RQ 3: What are the communication networks used to connect victims with connected contacts?
RM 3: To understand different networks and the technology used to connect victims and predefined numbers.

Table 3.1B
Explains different types of inputs and keywords.

Input	Keyword
Primary	Devices V Technology V IoT devices V wearables V Gadgets V Surveillance System V mobile app V Drone
Secondary	Women's safety V women's security
Tertiary	Physical violence V Rape V Threat V Street abusing V kidnapping V human trafficking
Narrow	India

analysis is conducted, and the table is compiled from various publications. Following the implementation of the SIRIP filter and limitations, the resulting list is presented below in Table 3.1E. Each column delineates the publication, author names, publication years, technology types, device initialization methods, deployment details, employment of ML algorithms, sensors utilized, and the outcomes or findings obtained from the respective research paper. The SIRIP analysis indicates whether each paper is included or not.

4. Result and analysis

The structured approach not only facilitates comparative analysis but also aids researchers, developers, and policymakers in identifying gaps, evaluating effectiveness, and designing context-specific interventions. Many safety apps and devices claim to enhance personal security; yet, there is limited empirical evidence on how they perform in real-life emergencies, particularly in diverse socio-economic and geographic settings across India.

Table 3.1C
Showing search thread applied in various sources.

Source	Search thread
IEEE Explore	("Devices OR Technology" OR "IoT devices" OR "wearables" OR "Gadgets" OR "Surveillance System" OR "mobile app" OR "Drone") AND ("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape" OR "Threat" OR "Street abusing" OR "kidnapping" OR "human trafficking") AND India
Research Gate	("Devices OR Technology" OR "Gadgets" OR "Surveillance System" OR "mobile app") AND ("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape") AND India
Science Direct	("Technology" OR "IoT devices" OR "wearables" OR "Gadgets" OR "Surveillance System" OR "mobile app") AND ("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape" OR "Threat" OR "Street abusing" OR "kidnapping")AND India
ACM Digital	("Devices OR Technology" OR "Surveillance System" OR "mobile app" OR "Drone") AND("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape" OR "Threat" OR "Street abusing" OR "human trafficking")AND India
Springer	("Devices OR Technology" OR "IoT devices" OR "wearables" OR "Gadgets" OR "Surveillance System" OR "mobile app" OR "Drone") AND("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape" OR "Threat" OR "Street abusing" OR "kidnapping" OR "human trafficking")AND India
AIP conference	("Devices OR Technology" OR "IoT devices")AND("Women's safety" OR "women's security") OR ("Physical violence" OR "Rape" OR "Threat" OR "Street abusing" OR "kidnapping" OR "human trafficking")AND India

Table 3.1D
SIRIP-filters and limitations strategy.

SI.no	Norms	Filters		Remarks
i.	Publication date (2018–2023)	>2018	YES NO	Excluded Included
ii.	Peer-reviewed articles	>20 citation	YES NO	Excluded Included
iii.	English language	YES NO		Included Excluded
Iv	Geographical focus	Within India	YES NO	Included Excluded
V	Clear Abstract	Yes NO		Included Excluded

4.1. A: General analysis of papers [Year-wise analysis]

A comprehensive review of papers is conducted for this study, encompassing a total of 78 papers.

4.2. A: Graph showing year-wise analysis

4.2.1. B. Publication

It outlines the diverse sources utilized and specifies the number of papers analyzed from each publication.

4.2.2. Inference

Notably, the study predominantly includes research papers published in 2022, with a majority sourced from IEEE Access or IEEE Explore.

4.3. A discussion on WST and different wearable devices

Various types of WSTs are employed to improve safety as well as security in public spaces. Integrating blockchain technology into women's safety technologies offers a promising solution for secure and tamper-proof data management, particularly in sensitive scenarios involving personal safety, real-time location tracking, and incident reporting. In the Indian context, where trust in institutional systems can be inconsistent, blockchain-based safety platforms can build user confidence and encourage broader adoption of digital safety tools among women (Rahman et al., 2022). Mobile applications have become a critical component of women's safety technologies, offering features like SOS alerts, real-time tracking, emergency contact reach-out, and distress signal activation, as documented in several studies (Akram et al., 2019; Chakrapani et al., 2022; Khandoker et al., 2019; Leema et al., 2021; Mareeswari & Patil, 2018; Opika & Rao, 2020; Vinarao et al., 2019). Smart wearable technologies, through biometric stress monitoring, analyze physiological biomarkers like Heart Rate Variability (HRV), ElectroDermal Activity (EDA), and Heart Rate (HR) to evaluate stress levels (Kallio et al., 2025). Table 4.2A explores the different types of

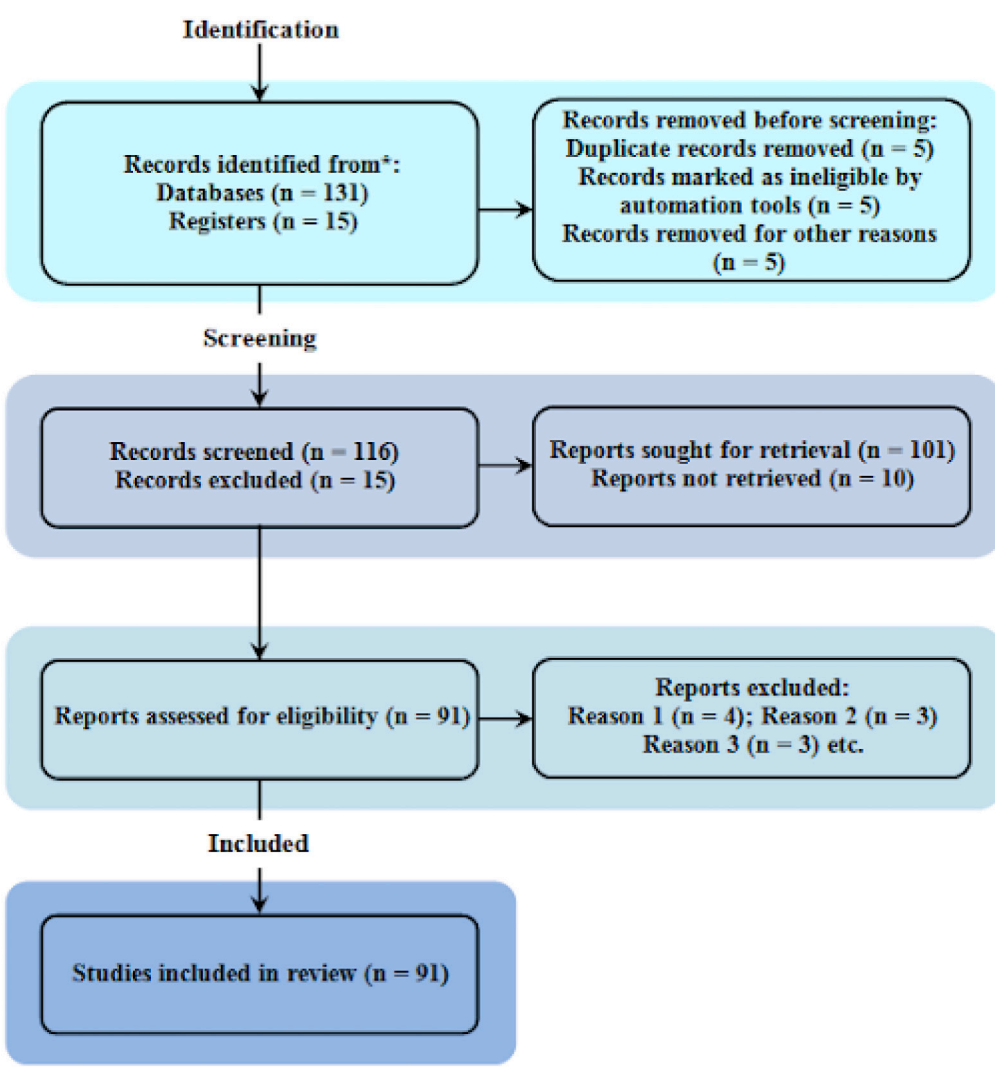

Fig. 2. PRISMA flowchart.

Table 3.1E
Analysis of papers.

Sl. no	Publication/ journal	Author	Deployed on	Activating mode	Communication mode	Usage of ML/ algorithm	Technology [H/w]	Type of sensor	Outcome	SIRIP analysis
1	IEEE- 2018	Muskan, T. Khandelw al *et.al*	Smartbandwatches	Automatically generates alertness by comparing normal and flicker thresholds.	Internet Protocol and Zigbee	ML algorithm (logistic regression)	Not specified	Temperature and heartbeat sensor	Generates an alarm and shares location with multiple contacts.	Included
2	IEEE- 2020(Yaswanth et al., 2020)	B. S. Yaswanth *et.al*	Smart bands, shoes, and watches	Panic button	Internet and wifi	IoT and KNN Algorithm	Raspberry pi	Fingerprint trained during a panic situation.	Share the user's location while they feel panic, and also aid in finding the nearest safe place. It also captures the culprit's photo and stores it in the cloud.	Included
3	IEEE- 2022(Chakrapani et al., 2022)	K. Chakrapa ni *et.al*	Wearable+android-based- phone app	Automatic sensors based on temp and pulse	GPS-GSM	NO	Microcontroller and Cloud-based	Temperature and pulse sensors	Alerts and provides location-based intelligence.	Included
4	IEEE- 2019(Vinarao et al., 2019)	E. D. G. Vinarao *et al.*,	Mobile-Based Application	Complaints are initiated by the users.	GPS- GSM and Active mobile network	NO	Android app	NO	Sends notifications *via* SMS, video, and image capturing and directly calls the police and GPS tracking in times of distress.	Included
5	IEEE- 2022(Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022)	S. P. Reddy *et al*	Wearable Handy Gadgets	Buzzer and Nerve simulator	GPS	IoT Technology	Not specified	Nerve stimulator	Tracks the location of women, informs the nearest police stations, and alerts the public by using a buzzer. The nerve stimulator generates an electric shock pulse, which helps to safeguard from criminals.	Included
6	IEEE- 2021(Tayal et al., 2021)	S. Tayal *et al*	Wearable Handy Gadgets	A buzzer and a push button	GSM and GPS	NO	Node MCU	NO	Tracks the woman's location and sends an emergency message to saved contacts and the nearby control room. Moreover, nearby people are alerted by the buzzer to help the women.	Included
7	IEEE- 2022(Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022)	KV *et al*	Wearable Handy Gadgets	push button	GSM	NO	Raspberry pi	NO	Share location, message, and call alert.	Exclude
8	IEEE- 2023	V. Gowrisha nkar *et.al*	Smart-Id Card	Automatic senses a loud noise	GPS	IoT	Cloud-based	Sound	Share the user's location and forward alerts to the suitable parties.	Included
9	IEEE- 2023(Gowrishankar et al., 2023)	U. Y. Reddy *et.al*	CCTV	Automatic Prediction of Weapons	NO	The Darknet 53 algorithm/ ML	Not specified	Prediction of weapons	Locate or find an alert for the situation for officers in the mall.	Included
10	IEEE- 2019(Sharma et al., 2019)	V. Sharma, Y. Tomar *et.al*	Embedded in a shoe [smart shoe]	Panic button	GPS and GSM module	NO	Raspberry Pi board and Arduino	NO	The location and image of the attacker shared with contacts generate shock.	Included
11	IEEE- 2022	N. Ananthula *et.al*	Wearable Handy Gadgets	Pressing the emergency button	GPS GSM modules	NO	Arduino and a microcontroller unit	NO	Shares location, message, and call alert	Included

(*continued on next page*)

Table 3.1E (*continued*)

Sl. no	Publication/ journal	Author	Deployed on	Activating mode	Communication mode	Usage of ML/ algorithm	Technology [H/w]	Type of sensor	Outcome	SIRIP analysis
12	IEEE- 2022(Saravanan et al., 2022)	K. A. Saravanan *et. al*	Sensor Paper can be attached to any Handy Gadgets.	Automatically detects the threshold.	GPS-GSM	NO	Paper	Pulse, tilt, and alcohol	Shares location, message, and call alert predefined numbers.	Included
13	IEEE- 2020(Dandamudi et al., 2020)	A. G. B. Dandamudi *et.al*	Drone	Trained to sense danger.	GPS Neo 6 m	CNN and ML algorithms	Raspberry Pi and Arduino connected with the Pi-camera.	Ultrasonic sensor	Alert the officers.	Included
14	AIP-conference 2022(Naved et al., 2022)	Naved, Mohd, and Fakih *et.al*	Wearable collar device with Android app	Sensors on a loud voice	GPS	YES	Raspberry pi	Pulse detectors and an accelerometer	When turning on the shouting warning, the electric shock device, the monitor, location details of contacts, and the closest police station are enforced for self-defense intent (video camera, electric shock, and streaming live footage).	Included
15	IEEE- 2022(Soundharya & Ashwini, 2022)	S. L. S *et.al*	Wearable Handy Gadgets	Button with self-defense tools	GSM and GPS	NO	Microcontroller and Cloud-based technology	NO	Send alert signals to neighbors and generate an electric shock. Additional features encompass audio recording and capturing the Culprit's image.	Included
16	IEEE- 2020(Agarkhed et al., 2020)	J. Agarkhed *et.al*	Wearable Handy Gadgets	Touch sensors and pulse sensors.	GPS-GSM	NO	Arduino	Touch sensor and pulse sensors	Shares location, message, and call alert for predefined numbers and generates shocks.	Included
17	IEEE 2019(Nagamma, 2019)	T. M. R, Aishwarya *et. al*	Wearable smart band	Button and sensor	GSM and GPS	IoT	Raspberry pi	Body posture while falling down and pulse rate.	Messages to predefined numbers.	Included
18	IEEE- 2020(Khan et al., 2020)	R. Khan, N. Mahfuz *et.al*	Wearable Handy Gadgets and a mobile app	Fingerprint sensor	GSM	IoT	Not specified	Fingerprint sensor, heartbeat, temp, and pulse.	Shares location with nearby users using the same app and sends messages to the preregistered number.	Included
19	IEEE- 2018(Sogi et al., 2018)	Sogi, P. Chatterjee *et. al*	Wearable smart ring	Emergency button	GSM	IoT	Raspberry Pi and Camera	NO	Shares the victim's location with a predefined number via the victim's phone and sends a photo of the attacker through the camera.	Included
20	IEEE- 2019(Khandoker et al., 2019)	R. R. Khandoker, S. Khondaker *et. al*	Mobile-Based Application	Voice message	GPS Active/ Offline mobile network	ML-algorithm	Not specified	YES	An alert message with the location is sent to the user-defined numbers every five minutes until the system is turned off.	Included
21	Springer- 2019(Mareeswari & Patil, 2018)	V. Mareeswari *et.al*	Android Mobile-Based Application	Review	GPS	NO	Not specified	NO	Message to pre-registered contact number.	Excluded
22	IEEE- 2021(Leema et al., 2021)	RJ leem *et. al*	Mobile-based applications and	By clicking the button.	GPS and Bluetooth	NO	ARM Interface	NO	Send a call to a registered number.	Included

(*continued on next page*)

Table 3.1E (*continued*)

Sl. no	Publication/ journal	Author	Deployed on	Activating mode	Communication mode	Usage of ML/ algorithm	Technology [H/w]	Type of sensor	Outcome	SIRIP analysis
			gadgets connected with Bluetooth							
23	Science- Direct-2023(Priya et al., 2023)	K. Shanmuga Priya [a], S. Vasanthi *et.al*	EEG and electrodes are put on the scalp	Blink of an eye	GPS	Naïve ML algorithm	Brain-Computer Interfaces (BCIs)-based on electroencephalograms (EEGs)	Pretrained blink eye sensors detect emotions like sad, happiness, and danger.	Sends notifications via SMS to predefined numbers and GPS tracking in times of distress.	Included
24	Science- Direct-2021(Ali & Goswami, 2023)	Fadai *et.a l*	Wearable Handy Gadgets	Emergency button and sensor	GPS and GSM	IoT	ARM processor and the Microcontroller (LPC812)	The temperature goes low.	Sends text messages like "help me" or "save me" for registered numbers.	Included
25	Research Gate-2019(Akram et al., 2019)	Wasim khan *et.al*	Mobile app	Fingerprint sensor	GPS and GSM modules	ML algorithm	Not specified	Fingerprint sensor	i) Send a message to a predefined number and generate shock waves ii) Send the location on the map to the receiver.	Included
26	IEEE- 2023(Farooq et al., 2023)	**M. S. Farooq, A. Masooma et.al**	All wearable handy gadgets	All	Review	All algorithms	**IoT devices**		Literature Review on IoT Devices.	Excluded
27	IEEE- 2021(Ali et al., 2021)	**Z. Ali, M.A. Khan, O. B. Samin et.al**	Gloves	Pulse Rate Sensor and emergency button	GPS and GSM	NO	**Arduino UNO and LCD**	**Pulse rate**	Sends alerts to predefined numbers.	Included
28	IEEE- 2022(Zadoo et al., 2023)	**M. Zadoo et al .,**	Surveillance system	Detecting danger	GPS	Not specified	**Not specified**	**NO**	Sends alerts and notifications to predefined numbers.	Included
29	Research Gate-2023(Harshitha et al., 2023)	**Harshitha, V. & Manusha, S. et.al**	Safety Devices	Pressing button	GPS and GSM	NO	**Arduino**	**NO**	Generates shock teasers and location tracking.	Included
30	ACM- 2022(Kaur et al., 2022)	**Jasmeeth Kaur et.al**	Embedded system	Detecting danger	GPS and GSM	NO	**Not specified**	**NO**	Sends alerts and notifications to predefined numbers.	Excluded
31	Springer2022 (Aqilah Arshad et al., 2022)	**Siti ramlaet. et.al**	Wearable handy gadgets	A motion detector sensor and a body temperature sensor	GPS	NO	**Not specified**	**Motion- detector, sensor, and body temperature re sensor**	Sends alerts and notifications to predefined numbers.	Included
32	Springer2022 (Mandal et al., 2023)	**G. Ranjithku Mar et.al**	Wearable gadgets	Automatically detects pulse rate.	GPS and GSM	NO	**Node MCU**	**Flex sensors and pulse sensors**	Alerts the nearby police station location and pre-registered contact number.	Included
33	Springer2021(Opika & Rao, 2020)	**V. Voorwashi et.al**	Mobile app- sensors	Emergency touch sensor	GPS, GSM, and wifi	IoT	**Not specified**	**Touch sensor**	Shares location, message, and call alert to predefined numbers.	Included

(*continued on next page*)

Table 3.1E (*continued*)

Sl. no	Publication/ journal	Author	Deployed on	Activating mode	Communication mode	Usage of ML/ algorithm	Technology [H/w]	Type of sensor	Outcome	SIRIP analysis
34	Springer2018 [41]	**Sunita.s patil et.al**	Mobile-based- Android app	By clicking the emergency button.	GPS	NO	**Not specified**	**NO**	An alert message is sent to predefined contacts.	Included
35	ReseachGate- 2020 [42]	**Ms.k.Opik a**	No	NO	NO	NO	**NO**	**NO**	Review paper	Excluded

available wearable devices. Advanced AI and deep learning algorithms are increasingly used to enhance Electroencephalogram (EEG) signal interpretation, improving accuracy in stress detection and identity verification (Salturk & Kahraman, 2025).

IoT devices have evolved as a key area of innovation in WST, with a total of 10 research papers focusing on IoT-based solutions (Ali & Goswami, 2023; Khan et al., 2020; Nagamma, 2019; Naved et al., 2022; Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022; Sogi et al., 2018; Yaswanth et al., 2020). However, challenges, such as affordability, battery life, connectivity, and user privacy, remain significant barriers to large-scale adoption (Bello & Figetakis, 2023). Surveillance systems, including CCTV cameras and video analytics, are widely deployed in public spaces to enhance monitoring and act as both deterrents and evidence-collection tools (Gowrishankar et al., 2023). Additionally, drones, or unmanned aerial vehicles, are increasingly utilized for real-time monitoring, emergency response, and surveillance of hard-to-reach or large areas (Dandamudi et al., 2020; Zadoo et al., 2023). Embedded systems contribute to a layered, technology-driven safety framework, particularly in high-risk or high-traffic zones (Kaur et al., 2022).

4.3.1. Inferences

Notably, 12 papers are dedicated to wearables, while 10 of these specifically address wearables integrated with IoT, summing up to a total of 22 wearable devices studied.

4.4. The prominent features included in developing these WSTs

A primary functionality of WSTs is real-time location tracking, enabled through the integration of GPS and location-based services, which allows for continuous monitoring and rapid response; an impressive 98 % of the gadgets and apps reviewed in this study include this feature. Another significant feature is audio and video recording, which can serve as essential evidence in the event of an assault. Approximately 30 % of the devices or applications include this capability (Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022).

Various sensors play a pivotal role in enhancing the functionality and responsiveness of women's safety technologies. Accelerometers are widely utilized for detecting sudden movements or changes in velocity, making them effective in identifying falls or abrupt motions (Naved et al., 2022). GPS sensors provide precise location data, facilitating real-time tracking and accurate positioning, which is a feature supported by several studies (Aqilah Arshad et al., 2022; Nagamma, 2019; Naved et al., 2022; Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022; Soundharya & Ashwini, 2022). For communication, GSM is commonly used, supporting cellular connectivity for calls, messages, and data transmission (Ali & Goswami, 2023; Chakrapani et al., 2022; Reddy,

Table 4.2A
Showing different types of available wearable devices.

Wearable devices/ gadgets	Reference No
Smart jewelry	(Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022)(Ananthula et al., 2022)(Saravanan et al., 2022)(Soundharya & Ashwini, 2022)(Agarkhed et al., 2020)(Sogi et al., 2018)(Ali & Goswami, 2023)(Farooq et al., 2023)(Aqilah Arshad et al., 2022)(Mandal et al., 2023)
Smart shoes	(Saravanan et al., 2022)(Sharma et al., 2019)
Smart watches or bands	(Ananthula et al., 2022)(Saravanan et al., 2022)(Tayal et al., 2021)(Soundharya & Ashwini, 2022)(Agarkhed et al., 2020)(Nagamma, 2019)(Ali & Goswami, 2023)(Farooq et al., 2023)(Harshitha et al., 2023)(Aqilah Arshad et al., 2022)(Mandal et al., 2023)
Smart paper	(Saravanan et al., 2022)
Smart clothing/ Gloves	(Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022)(Naved et al., 2022)
EEG	(Priya et al., 2023)

Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022; Sogi et al., 2018). Additionally, HR monitors help detect abnormal HR variations, which may indicate physical stress or emotional distress, and are employed across various studies (Agarkhed et al., 2020; Akram et al., 2019; Khandelwal et al., 2018; Naved et al., 2022; Reddy, Konguvel, et al., 2022; Reddy, Vamsi, et al., 2022; Saravanan et al., 2022). Camera and image sensors are employed in surveillance and wearable devices to capture images or record video, serving both monitoring and evidentiary functions (Gowrishankar et al., 2023). Temperature sensors monitor both environmental and body temperature changes, helping detect abnormal physical conditions (Chakrapani et al., 2022;Khan et al., 2020; Khandelwal et al., 2018). Lastly, biometric sensors, including fingerprint scanners, iris recognition, and facial recognition technologies, are used for secure authentication or controlled access, as demonstrated in multiple studies (Agarkhed et al., 2020; Khan et al., 2020; Khandoker et al., 2019; Yaswanth et al., 2020).

ML algorithms play a noteworthy role in advancing women's safety technologies by enabling predictive analysis, threat detection, and pattern recognition. Commonly implemented algorithms identified in this study include logistic regression, darknet, and Convolutional Neural Networks (CNN), which make a contribution to enhancing the accuracy and responsiveness of safety systems (Akram et al., 2019; Farooq et al., 2023; Gowrishankar et al., 2023; Khandelwal et al., 2018). Synchronizing with mobile apps provides users with seamless control, real-time alerts, and additional features that improve user experience and safety management (Akram et al., 2019; Dandamudi et al., 2020; Gowrishankar et al., 2023; Khandelwal et al., 2018; Priya et al., 2023; Yaswanth et al., 2020).

4.4.1. Inference

Approximately 30 % of the papers incorporate ML algorithms or sensors, while 20 % of the studies document audio/video recording capabilities and self-defense mechanisms.

4.5. The communication networks implemented to connect the designated emergency contacts

WSTs leverage various communication networks to enhance connectivity and ensure timely response during emergencies. Cellular networks (2G/3G/4G/5G) utilize existing mobile infrastructure to make calls, send text messages, and transmit data to predefined emergency contacts. These networks incorporate GPS technology for precise location tracking and GSM as the communication standard that facilitates voice, text, and data transmission, thereby enabling simultaneous communication and location tracking to improve safety features (Agarkhed et al., 2020; Saravanan et al., 2022). Within this context, Zigbee, a wireless communication protocol centered on the IEEE 802.15.4 standard, supports low-power, short-range communication, making it ideal for energy-efficient safety applications (Khan et al., 2020; Khandelwal et al., 2018). Furthermore, Bluetooth technology is widely used for short-range communication between safety devices and paired smartphones, enabling quick alerts and seamless data transfer (Leema et al., 2021).

4.5.1. Inference

Interestingly, among 78 papers reviewed, only one paper emphasizes the use of Zigbee networks as a contingency measure in case of network failure.

5. Shortcomings of WST, a discussion, and proposed taxonomy

The analysis of the papers reveals their individual shortcomings. Muskan et al.'s research indicates that the development of these devices is proven to be quite expensive. Similarly, according to references (Gowrishankar et al., 2023) and (Dandamudi et al., 2020), the accessibility of drones and surveillance networks during emergencies is limited, as flaws are only identified post-crime. Additionally, reliance on the internet or encountering no network coverage can compromise the required level of protection (Vinarao et al., 2019) and (Gowrishankar et al., 2023).

5.1. Proposed taxonomy

Based on these findings, a summarized overview has been generated. Fig. 3 explores the proposed taxonomy. An architecture has been suggested, delineating various subcategories of technologies used, along with their features or outlines, as presented below.

Four-tier taxonomy is proposed, classifying entities based on types, subcategories, technologies, and features. Technology specifies the use of sensors, GPS/GSM, algorithms, wearables, or other features. The review also emphasizes the significance of contextual relevance, affordability, and accessibility, all of which have a direct impact on the development and application of future technologies.

6. Research gaps

Although this work examines 78 WST-related studies, it is crucial to specifically point out the gaps in the current literature in order to enhance its scholarly value. First, thorough evaluations that critically assess and comprehensively classify WST across different technological fields are conspicuously lacking. Second, there is little scholarly attention that is especially suited to the Indian setting, despite the country's distinct sociocultural terrain, inequalities in infrastructure, and high prevalence of gender-based violence. Third, not enough attention is paid to user-centric factors like pricing, accessibility, usability, and practical adoption of these technologies, especially among women from underrepresented groups. Lastly, there is a need for more interdisciplinary research that bridges technology development with cultural, behavioral, and policy dimensions to create more effective, scalable, and context-sensitive safety solutions. India also poses a distinct set of difficulties that have a big influence on the advancement and application of WST. There is a pressing need for WST solutions that are affordable, scalable, and tailored to India's socio-economic diversity to ensure meaningful impact and widespread adoption, especially among women in underserved and vulnerable communities.

7. Conclusion

The field of WST saw a growing body of research, primarily focusing on wearable devices, gadgets, and mobile applications. This paper applied the SIRIP framework to systematically review literature on WST with an Indian perspective. From an initial pool of 36 publications, 32 were selected based on relevance and citation strength. By systematically analyzing and categorizing various technologies, such as mobile applications, the findings guided the development of more inclusive, accessible, and culturally relevant safety solutions, especially for women in marginalized or rural communities. The study also highlighted a significant gap in user awareness and real-world adoption, indicating that despite the availability of various tools, few women actively used them in daily life. Women's safety technologies, ranging from personal alarms and GPS-enabled devices to mobile applications and self-defense tools, were vital in promoting autonomy and security. This review contributed to the literature by presenting a structured synthesis of existing technologies and identifying key areas for further innovation and policy focus. WST's future depended on how well it integrated ethical, cultural, and institutional dimensions alongside technological progress.

7.1. Ethical complications

The ethical implications of WST are a critical yet often underexplored dimension of the field, warranting careful consideration

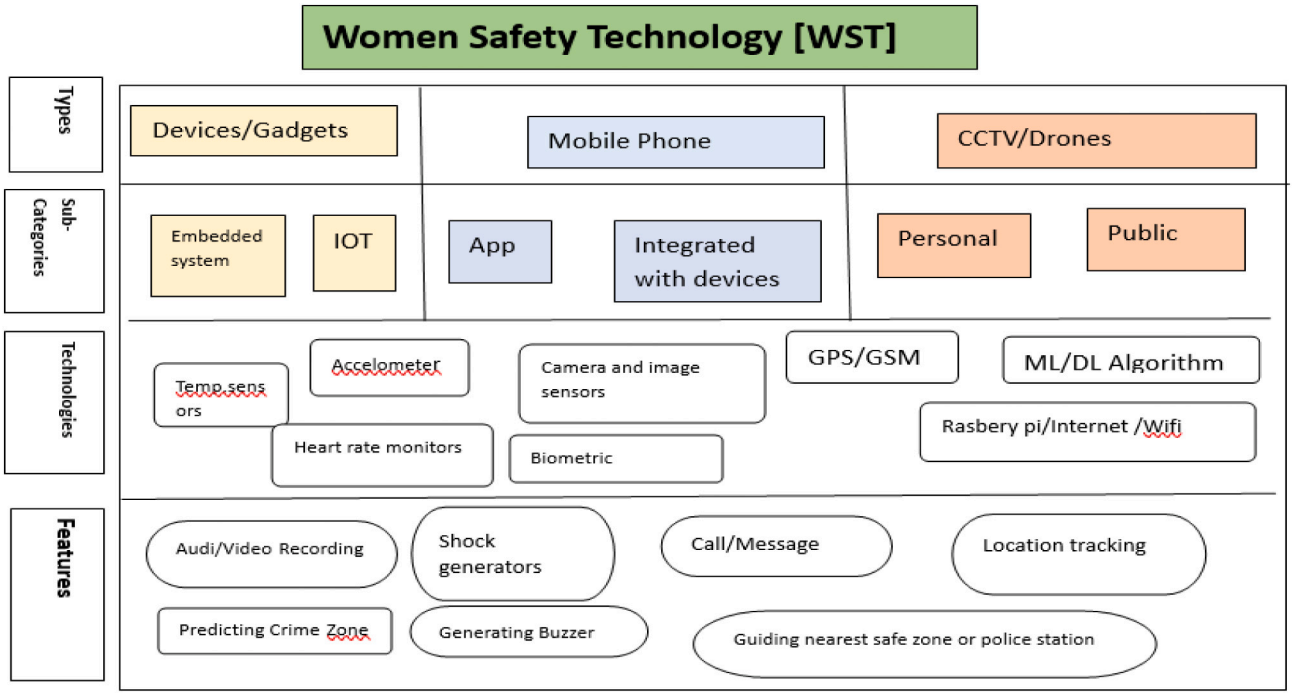

Fig. 3. Proposed taxonomy.

alongside technical advancement. While WST solutions, such as GPS tracking, surveillance systems, and wearable alert devices, offer significant benefits in enhancing personal security and enabling timely interventions, they also raise concerns around privacy, data security, informed consent, and potential misuse. Therefore, this review not only assesses the effectiveness and accessibility of WST but also aims to contribute to the formulation of ethical guidelines by encouraging a balanced discourse on both its promises and risks.

CRediT authorship contribution statement

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Consent of publication

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Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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